

Final Project: Drug Interaction Informant

Emily Youngward

Zoe Mann

Our Project and Use Case

- The purpose of our project is to allow users to ask our model whether or not they can safely take two different medications together. Our model then responds to the user's query by answering whether the medication combo is safe, along with any potential medication interactions.
- The use cases would include anyone curious about starting a new medication, individuals taking over the counter medications they are unsure about, and individuals who want to double check their current prescribed medications to understand any potential interactions better.

The Dataset

- The dataset we are using comes from Kaggle.

- It includes three columns:

Column 1 = First drug name

Column 2 = Second drug name

Column 3 = Drug Interaction description

The set contains more than 1600 unique drugs

Abemaciclib	Sulfisoxazole	The metabolism of Sulfisoxazole can be decreased when combined with Abemaciclib.
-------------	---------------	--

- Drug-Drug Interaction Dataset:
<https://www.kaggle.com/datasets/mghobashy/drug-drug-interactions>

Bias/Dataset Issues

- Due to the dataset only featuring generic drug names we had to find a way to have the model understand when the user is asking about brand name drugs.
- We concluded that since this model relies on RAG, and thus tends to match the generic drug name to the medical drug name on its own, we did not have to manually match these ourselves.
- We had to reformat the data set to work for our model. We formatted the data so that each row was a separate document. We then extracted all the drugs into a drug list so that we could check whether or not a users query contained any drugs in that drug list.

Model Used

- This project utilizes RAG with LLAMA.
- We experimented with using both GPT2 and LLAMA, and overall we found LLAMA gave us more accurate results.
- The model may be biased towards North American drugs due to our dataset and the fact that LLAMA is made by Meta, a American company.

Guardrails

- This guardrail provides a disclaimer at the end of each response:
 - “This chatbot should not be used in replacement of a licenced medical practitioner, as it may contain inaccuracies. Please always consult your medical practitioner before starting and/or stopping any medications.”
- This guardrail asks users to re-enter two drug names as a query if no medications are detected:
 - “Please re-enter two drug names so that I can check whether or not they can be safely taken together. If you had entered two drug names, I could not find information about this drug combination.”

Initial Results - testing GPT2

These are the initial results with GPT2.

Query: Cyamemazine and verteporfin?

Top Retrieved Documents:

		Document	Similarity
4	Cyamemazine	Verteporfin Cyamemazine may increa...	0.758941
126121	Venlafaxine	Cyamemazine The risk or severity o...	0.672516

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

Both `max\_new\_tokens` (=256) and `max\_length` (=150) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. ([https://huggingface.co/openai/gpt2](#))

Response from LLM with RAG: Respond the query based on the context. Context: Cyamemazine Verteporfin Cyamemazine may increase the photosensitizing activities of Verteporfin..

Venlafaxine Cyamemazine The risk or severity of adverse effects can be increased when Venlafaxine is combined with Cyamemazine.. Query: Cyamemazine and verteporfin?

Venlafaxine (Dry) Cyamemazine The risk or severity of adverse effects can be increased when Venlafaxine is combined with Cyamemazine.. Query: Cyamemazine and verteporfin?

Venlafaxine (Spice) Cyamemazine or Venlafaxine may increase the photosensitizing activities of Spice. Query: Cyamemazine and Spice?

Venlafaxine (Sunscreen) Cyamemazine or Venlafaxine may increase the photosensitizing activities of Sunscreen. Query: Cyamemazine and Sunscreen?

Venlafaxine (Zinc) Cyamemazine or Venlafaxine may increase the photosensitizing activities of zinc. Query: Cyamemazine and zinc?

Venlafaxine (Other) Cyamemazine or Venlafaxine may increase the photosensitizing activities of other antioxidants. Query: Cyamemazine and other antioxidants?

Methylmercury (Methylmercury) Cyamemazine could increase the photosensitizing activities of methylmercury. Query: Cyamemazine and methylmercury?

These were the initial results with GPT2. The GPT2 model did not do a good job of responding to the query in an understandable manner. It also continued to go off-topic.

Initial Results

With Llama

Query:

I want to see the movie Wicked.

Response:

“Response from LLM with RAG: Please re-enter two drug names so that I can check whether or not they can be safely taken together. If you had entered two drug names, I could not find information about this drug combination.”

```
query = "I want to see the movie Wicked."  
response = retrieve_with_pandas(query, top_k=2)  
  
print("\nQuery:", query)  
print("\nTop Retrieved Documents:")  
print(response)
```

Query: I want to see the movie Wicked.

Top Retrieved Documents:

	Document	Similarity
185888	Isocarboxazid Tryptophan Isocarboxazid may inc...	0.203383
185931	Levomilnacipran Tryptophan Levomilnacipran may...	0.202262

Initial Results

With Llama

Query:

Can I take bortezomib and digoxin?

Response:

“Bortezomib may decrease the cardiotoxic activities of Digoxin, which means that taking Bortezomib and Digoxin together may reduce the risk of cardiotoxicity associated with Digoxin. However, it is important to note that this interaction is only significant in certain situations, such as when Digoxin is given at high doses or in combination with other medications that can also decrease its metabolism. In general, the combination of Bortezomib and Digoxin is considered safe and effective, but it is important to consult with a healthcare professional to determine the appropriate dosage and monitoring regimen. This chatbot should not be used in replacement of a licenced medical practitioner, as it may contain inaccuracies. Please always consult your medical practitioner before starting and/or stopping any medications.”

```
# Ask the same question with RAG
response_rag = generate_with_rag(query)
print("****")
print("Response from LLM with RAG:", response_rag)
```

Query: Can I take Bortezomib and Digoxin? Answer: Bortezomib may decrease the cardiotoxic activities of Digoxin, which means that taking Bortezomib and

```
query = "Can I take Bortezomib and Digoxin?"
response = retrieve_with_pandas(query, top_k=2)
```

```
print("\nQuery:", query)
print("\nTop Retrieved Documents:")
print(response)
```

Query: Can I take Bortezomib and Digoxin?

Top Retrieved Documents:

	Document	Similarity
55	Bortezomib Digoxin Bortezomib may decrease the...	0.806928
43667	Digoxin Voriconazole The metabolism of Voricon...	0.702608

Initial Results

With Llama

Query:

Can I take Prozac and Zoloft?

```
# Ask the same question with RAG
response_rag = generate_with_rag(query)
print("****")
print("Response from LLM with RAG:", response_rag)
```

Can I take Prozac and Zoloft? Answer: It is not recommended to take Prozac (fluoxetine) and Zoloft (sertraline) together, as this can increase the risk

Response:

“It is not recommended to take Prozac (fluoxetine) and Zoloft (sertraline) together, as this can increase the risk of adverse effects. Both Prozac and Zoloft are selective serotonin reuptake inhibitors (SSRIs), which work by increasing the levels of the neurotransmitter serotonin in the brain. Taking two SSRIs together can increase the risk of serotonin syndrome, a potentially life-threatening condition caused by excessive levels of serotonin in the body. Additionally, combining Prozac and Zoloft may increase the risk of side effects such as dizziness, drowsiness, and nausea. If you are taking either of these medications and are considering adding the other, it is important to consult with your healthcare provider to determine the best course of action. They can help you weigh the potential benefits and risks. This chatbot should not be used in replacement of a licenced medical practitioner, as it may contain inaccuracies. Please always consult your medical practitioner before starting and/or stopping any medications.”

```
query = "Can I take Prozac and Zoloft?"
response = retrieve_with_pandas(query, top_k=2)
```

```
print("\nQuery:", query)
print("\nTop Retrieved Documents:")
print(response)
```

Query: Can I take Prozac and Zoloft?

Top Retrieved Documents:

	Document	Similarity
124192	Procarbazine Zolopine The risk or severity of ...	0.617265
78990	Sertraline Procarbazine The risk or severity o...	0.607736

Improvements for Next Week

- We would like to work on the model's recognition of brand and generic names of drugs. We asked the model about certain brand names, in this case, Prozac and Lexapro (SSRIs). The model could not provide an accurate answer because the top retrieved documents did not contain a single matching generic to either Lexapro or Prozac. Thus, the context being provided to RAG was not useful for searching its external database.

Example:

```
Based on the context you provided, it is not recommended to take Prozac (Fluoxetine) and Lexapro (Escitalopram) together. The combination of Carbamazepine and Procarbazine or Procaine can increase the risk or severity of adverse effects.
```

```
query = "Can I take Prozac and Lexapro?"
response = retrieve_with_pandas(query, top_k=2)
```

```
print("\nQuery:", query)
print("\nTop Retrieved Documents:")
print(response)
```

Query: Can I take Prozac and Lexapro?

Top Retrieved Documents:

	Document	Similarity
78888	Carbamazepine Procarbazine The risk or severit...	0.608861
146017	Carbamazepine Procaine The risk or severity of...	0.599321

Circumstances of Success and Failure

- Due to the consequences surrounding the medical nature of the model, we have to implement extensive guardrails that inform the user the model is not official medical advice. Thus, the model is limited to being a supplementary companion.
- We have to consider where user's queries are being stored. This is because users may be disclosing personal medical information within queries.
- The model however, will succeed at its intended purpose of being an informant that may help trigger a meaningful conversation with a medical professional.

Thank you!

